ADVANCED PROGRAMMING WITH PROJECT-I/MINOR PROJECT REPORT

Entitled

A.L.E.R.T. (AUTOMATED LANDMARK-BASED EYE AND RESPONSE TRACKER)

Submitted by

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CERTIFICATE

This is to certify that Kavya Kothari (2301030600045), Prem Deshani (2301030600043) and Mehulsinh Rathod (2301030600049) have submitted the Advanced Programming with Project-I/Minor Project report entitled A.L.E.R.T. (Adaptive Landmark Eye-state & Reaction-Time) in partial fulfillment of the requirement for the award of the degree in Bachelor of Technology in Computer Engineering (AI & ML) of Silver Oak University, Ahmedabad during the academic year 2026-2027.

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DECLARATION

We hereby declare that the Advanced Programming with Project-I/Minor Project report submitted along with the Project entitled A.L.E.R.T. (Automated Landmark-based Eye and Response Tracker) in partial fulfillment for the Bachelor of Technology in Computer Engineering (AI & ML) to Silver Oak University, Ahmedabad, is a bonafide record of original work carried out by us at the College of Technology under the supervision of Ms. Vishwa Patel.

We further declare that the report does not contain any material previously submitted for the award of any degree or diploma in this or any other university. It is also certified that no part of this report has been copied from any source without proper acknowledgment, and all referenced materials, figures, and information from other sources have been appropriately cited in the report.

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ABSTRACT

Driver fatigue and micro-sleep lapses are implicated in over 20% of fatal vehicular collisions globally. Camera-based Driver Monitoring Systems (DMS) are non-intrusive, but existing solutions suffer from three architectural flaws: latency and privacy risks of cloud streaming; train-serve skew caused by natural anatomical eye variance across drivers; and mode collapse on naturalistic datasets caused by symmetric loss penalties over an intrinsically ordinal biological spectrum.

This project delivers **ALERT (Automated Landmark-based Eye and Response Tracker)**, a fully offline, edge-compliant driver fatigue monitor for real-time monocular deployment on consumer automotive CPUs. ALERT captures video from a standard webcam (640×480 @ 15 FPS) and extracts 478 3D facial landmarks with the MediaPipe Face Landmarker. It synthesizes a 12-dimensional kinematic feature vector ($F = 12$) of normalized aperture ratios, differential velocities (ΔEAR, ΔMAR, ΔPitch) and event metrics over a 120-frame (8.0 s) window.

To remove the anatomical bias arising from the observed 0.1800 to 0.4500 ($2.50\times$) spread in resting eye aspect ratio across subjects, ALERT performs an automated 15-second personal baseline calibration ($\mathrm{EAR}/\mathrm{EAR}_{\mathrm{base}}$). To overcome the central attractor collapse seen in early training, an **Ordinal Distance-Penalized Focal Loss** ($\lambda_{\mathrm{ord}} = 0.35$, $\gamma = 2.0$, $\epsilon = 0.08$) was formulated to penalize safety-critical opposite-extreme errors quadratically.

On the naturalistic **UTA-RLDD** benchmark (60 subjects, 141 videos, 26,050 calibrated windows), a 9-model heterogeneous ensemble of 1D temporal convolutions with GRU (HybridCNNGRU), dilated causal residual networks (TCNClassifier) and bidirectional GRUs with multi-head self-attention (BiGRUAttention) achieved a peak subject-independent holdout Macro-F1 of **0.6062** (accuracy **60.37%**) and an ensemble average Macro-F1 of **0.5236**, above the published subject-independent baselines. The deployable HybridCNNGRU reaches a single-thread CPU latency of **15.30 ms** (65.3 FPS) with a 0.069 MB checkpoint; the full 9-model ensemble occupies **2.35 MB** on disk. Real-time webcam testing shows zero false-positive alarms on awake drivers.

Keywords: Driver Fatigue Detection, Computer Vision, MediaPipe, Eye Aspect Ratio, PERCLOS, Heterogeneous Ensemble, Edge AI, Ordinal Focal Loss, On-Device Inference.

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LIST OF ABBREVIATIONS

Abbreviation	Expanded Form
ALERT	Automated Landmark-based Eye and Response Tracker
AMP	Automatic Mixed Precision
BiGRU	Bidirectional Gated Recurrent Unit
CAN	Controller Area Network
CLAHE	Contrast-Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DDAW	Driver Drowsiness and Attention Warning
DMS	Driver Monitoring System
EAR	Eye Aspect Ratio
ECG	Electrocardiogram
EEG	Electroencephalogram
EMA	Exponential Moving Average
ERT	Ensemble of Regression Trees
FPS	Frames Per Second
GPU	Graphics Processing Unit
HUD	Heads-Up Display
MAR	Mouth Aspect Ratio
MOE	Mouth-Over-Eye Ratio (MAR/EAR)
NHTSA	National Highway Traffic Safety Administration
NTHU-DDD	National Tsing Hua University Driver Drowsiness Detection Dataset
ONNX	Open Neural Network Exchange
PERCLOS	Percentage of Eye Closure
PnP	Perspective-n-Point
PPG	Photoplethysmography
PVT	Psychomotor Vigilance Task
RLDD	Real-Life Drowsiness Dataset (University of Texas at Arlington)
SoC	System on Chip
TCN	Temporal Convolutional Network

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CHAPTER 1 INTRODUCTION

1.1 Project Summary

ALERT (Automated Landmark-based Eye and Response Tracker) is an edge-native, monocular computer-vision DMS that detects cognitive fatigue, yawning and micro-sleep events in real time on resource-constrained consumer CPUs. It processes raw RGB video without infrared hardware or cloud offloading and computes a 12-channel kinematic feature vector over a 120-frame (8.0 s) window. An automated 15-second personal calibration establishes driver-specific facial geometry and prevents false alarms caused by anatomical differences. A heterogeneous 9-model deep ensemble, trained with a custom Ordinal Distance-Penalized Focal Loss, classifies fatigue into four actionable alert tiers.

1.2 Background and Motivation

NHTSA documents over 91,000 police-reported fatigue collisions annually in the United States, with 50,000 severe injuries and 800 fatalities [1]. A 2024 AAA Foundation study found that 17.6% of fatal crashes involved drowsy driving [2]. At 100 km/h a vehicle covers 27.8 m/s, so a 2-second micro-sleep leaves it unpiloted for 55.6 m.

Regulators have responded. Regulation (EU) 2019/2144 requires Driver Drowsiness and Attention Warning (DDAW) systems in all newly registered vehicles from July 2024 [3], and Euro NCAP 2026 mandates direct ocular monitoring for maximum safety ratings.

Existing approaches have clear drawbacks:

1. *Vehicle telemetry* (steering torque, lane departure) is reactive: alarms sound only after the vehicle trajectory has degraded [4].
2. *Physiological wearables* (EEG caps, chest harnesses) are clinically sound but intrusive and unviable for commercial drivers.
3. *Cloud-based vision systems* suffer network latency ($100 - 500$ ms), fail in connectivity dead zones and create biometric privacy risks under GDPR/CCPA.

ALERT addresses this gap with non-intrusive optical analysis running entirely on local automotive-grade processors.

1.3 Purpose

The purpose of ALERT is to develop, validate and package a lightweight, high-precision fatigue monitoring pipeline that achieves strong detection on a naturalistic benchmark while meeting automotive edge budgets for the deployed model: single-thread CPU execution under 30 ms, fewer than 250k parameters, and absolute biometric privacy.

1.4 Objectives

1. **Fully offline pipeline**: remove all cloud and network dependencies through edge inference on consumer x86/ARM CPUs.
2. **Kinematic signal synthesis**: augment static geometric metrics with velocity differentials (ΔEAR, ΔMAR, ΔPitch) over a 120-frame window.

3. **Anatomical baseline calibration**: implement an automated 15-second protocol that sets individual thresholds ($0.75 \cdot \mathrm{EAR}_{\mathrm{base}}$) and resolves the $2.50\times$ spread in human eye aperture.
4. **Ordinal loss formulation**: design a loss that penalizes opposite-extreme errors $4\times$ harder than benign boundary errors.
5. **Heterogeneous deep ensemble**: train and benchmark a 9-model ensemble across three architecture families (1D-CNN+GRU, dilated TCN, BiGRU+Attention) on UTA-RLDD under strict subject-independent cross-validation.
6. **Real-time edge packaging**: verify zero-false-positive live inference, pass the automated integration tests and provide Linux shell launchers.

1.5 Scope

The implementation covers monocular RGB video (640×480 @ $15-30$ FPS), landmark tracking under standard lighting, cross-validation across 60 subjects of UTA-RLDD, and live desktop/in-vehicle prototype execution. Infrared-only night conditions and active haptic CAN-bus actuation are left to commercial deployment.

CHAPTER 2 LITERATURE REVIEW

2.1 Technology and Related Work

2.1.1 Eye Closure Measures (EAR and PERCLOS)

Soukupová and Čech [5] defined the Eye Aspect Ratio (EAR) from six landmarks per eye:

$$\text{EAR} = \frac{\| p_2 - p_6 \| + \| p_3 - p_5 \|}{2 \cdot \| p_1 - p_4 \|}$$

EAR is an instantaneous proxy for eyelid distance normalized by eye width. Wierwille et al. [4] established PERCLOS (percentage of eyelid closure over the pupil $\geq 80\%$) in driving-simulator studies, later validated by Dinges et al. [6], [7] against Psychomotor Vigilance Task (PVT) lapses. NHTSA accepts PERCLOS as the gold standard for ocular drowsiness, but Abe [8] showed that static PERCLOS thresholds miss fluctuating early fatigue and ignore physiological differences between drivers.

2.1.2 Facial Landmark Extraction Pipelines

Early vision-based DMS used Viola-Jones Haar cascades and the Ensemble of Regression Trees (ERT) of Kazemi and Sullivan [9], implemented in dlib (68 landmarks). dlib incurs CPU latency of $30 - 60$ ms and degrades rapidly beyond $20°$ of yaw.

Lugaresi et al. [10] and Kartynnik et al. [11] introduced MediaPipe Face Mesh, which infers dense 3D surface geometry with a two-stage lightweight neural cascade. The MediaPipe Face Landmarker task [12] predicts 478 3D landmarks in under 10 ms on mobile/CPU hardware through the TensorFlow Lite XNNPACK delegate, giving the temporal precision ALERT requires.

2.1.3 Temporal Sequence Modeling

Gated Recurrent Units [17] model dependencies across sequential feature vectors. Additive attention [18] lets a model weight critical micro-sleep frames instead of relying on the final hidden state. Temporal Convolutional Networks [19] use causal dilated convolutions to span long receptive fields without recurrent state bottlenecks.

2.2 Datasets and Reported Results

1. **NTHU-DDD** [13]: 36 subjects in a driving simulator under infrared illumination with acted drowsiness. Accuracies above 73% are reported [14], [15], but acted fatigue does not represent genuine cognitive collapse.
2. **UTA-RLDD** [14]: the standard naturalistic drowsiness dataset, with about 30 hours of multi-angle video of 60 subjects in genuine alert, low-vigilant and sleep-deprived states. Its authors report a blink baseline of 65.2% video-level accuracy against 57.8% for human observers [14].

Table 2.1: Public fatigue datasets and reported literature baselines

Dataset	Modality	Genuine / Acted	Subjects	Validation Split	Published Result
NTHU-DDD [13]	Infrared	Acted	36	Subject-disjoint	73.06% Acc [15]
UTA-RLDD [14]	RGB video	Genuine	60	Subject-disjoint baseline	65.20% video Acc [14]

Dataset	Modality	Genuine / Acted	Subjects	Validation Split	Published Result
UTA-RLDD [16]	RGB video	Genuine	60	Random frame split (leakage)	86.52% Acc [16]
UTA-RLDD (ALERT)	RGB video	Genuine	60	Strict subject-independent (GroupKFold)	**0.6062** peak F1 / **0.5236** ensemble F1

2.3 Research Gap and Limitations of Prior Work

3. **Evaluation leakage.** Works reporting above $85-90\%$ on UTA-RLDD use random frame splitting, so consecutive frames of one subject appear in both training and test sets and networks memorize facial morphology instead of fatigue features. Under strict subject-independent GroupKFold the realistic ceiling on UTA-RLDD is $52-58\%$ **Macro-F1** [14].
4. **The circadian vs. instantaneous labeling paradox.** UTA-RLDD labels are assigned per 10-minute session from prior sleep deprivation. In a Low-Vigilant video the eyes may stay wide open for 3 minutes before a 3-second micro-sleep, so standard Cross-Entropy heavily penalizes correct Alert predictions and collapses the model into the intermediate class.
5. **Anatomical baseline blindness.** Prior work uses fixed population thresholds ($\text{EAR} < 0.25$), but resting eye aperture in our audit varied from 0.18 to 0.45 across subjects ($2.50\times$), which invalidates any static threshold.

CHAPTER 3 PROJECT PLANNING AND SCHEDULING

3.1 DEVELOPMENT APPROACH AND JUSTIFICATION

Development followed five empirical phases:

1. **Phase 1**: baseline modular pipeline (MediaPipe, geometric features, baseline BiGRU).
2. **Phase 2**: candidate multi-model edge architectures and identification of training bottlenecks.
3. **Phase 3**: cross-validation, root-cause analysis of validation loss divergence, and the master model.
4. **Phase 4**: 12-channel kinematic overhaul, personal baseline calibration, Ordinal Distance-Penalized Focal Loss and the heterogeneous ensemble.
5. **Phase 5**: end-to-end integration, the 57-test automated suite and live webcam stress-testing.

3.2 Team Organization and Responsibilities

Project ID: CE_AIML_31. **Department**: Computer Engineering (AI & ML), College of Technology, Silver Oak University. **Project Guide**: Ms. Vishwa Patel, Assistant Professor.

Member	Enrolment No.	Modules and Responsibilities
Kavya Kothari	2301030600045	Facial landmarking pipeline (MediaPipe Tasks), head-pose estimation (solvePnP), geometric features (EAR, MAR, MOE).
Prem Deshani	2301030600043	Model architectures (HybridCNNGRU, TCNClassifier, BiGRUAttention), Ordinal Focal Loss, edge latency benchmarks, Bash/Linux launchers.
Mehulsinh Rathod	2301030600049	High-throughput video extraction (cap.grab()), personal baseline calibration (Calibrator), soft-voting ensemble, online EMA filter, debounced alert state machine.

3.3 Schedule, Milestones and Periodic Review Summary

Table 3.1: Periodic review summary and milestones

Review	Interval	Work Completed	Progress
Review 1	08/08/2026 – 19/09/2026	Literature review, MediaPipe Tasks setup, baseline 8-channel geometric extractor, initial BiGRU, live HUD overlay.	80%
Review 2	20/09/2026 – 26/09/2026	Decoding of 141 UTA-RLDD videos, feature cache (84,942 windows), augmentation fixes, CPU training baseline.	90%
Review 3	27/09/2026 – 24/10/2026	Resolution of attractor collapse and train-serve skew; 12 kinematic channels; personal calibration; Ordinal Distance-Penalized Focal Loss; 9-model ensemble training; latency benchmarking; 57/57 integration tests; live webcam verification.	**100%**

CHAPTER 4 SYSTEM ANALYSIS

4..1 Study of Current Systems

Industrial DMS deployments fall into two groups:

7. *Proprietary in-cabin Tier-1 hardware* (e.g., Seeing Machines, Smart Eye): near-infrared illumination arrays and custom ASICs, too costly for retrofitting budget vehicles.
8. *Cloud-connected fleet dashcams*: video is sent over LTE/5G for remote inference. In rural corridors or tunnels, delays exceed 500 ms or the link drops, which defeats accident prevention.

4.2 Problems and Vulnerabilities in Existing Solutions

- **Train-serve skew**: models trained with static thresholds fail on live cameras when lighting or face distance changes.
- **Biometric exposure**: transmitting facial imagery over wireless networks creates privacy risk under GDPR, CCPA and the Indian Digital Personal Data Protection (DPDP) Act.
- **Binary alarm hysteresis**: simple threshold triggers beep during voluntary mirror glances, annoying drivers until they disable the system.

4.3 Requirements Specification of the Proposed System

Table 4.1: Edge DMS requirements specification

Category	Requirement	Target
Functional	Real-time landmark ingestion	478 3D landmarks via MediaPipe Face Landmarker
Functional	Personal baseline calibration	15-second open-eye session setting $T_{\text{EAR}} = 0.75 \cdot \text{EAR}_{\text{base}}$
Functional	Multi-tier fatigue categorization	3-class probability fusion into 4 debounced states
Functional	Visual and audio alerts	Debounced chime (Warning) and siren (Critical)
Non-functional	On-device latency (deployed single model)	Single-thread CPU mean latency ≤ 30.0 ms (≥ 33.3 FPS)
Non-functional	Model footprint	Deployed model < 250k parameters; checkpoint storage < 2.50 MB
Non-functional	Biometric privacy	No cloud transmission; no raw facial imagery stored
Non-functional	Runtime memory	Process RAM $< 1{,}000$ MB

4.4 System Feasibility

- **Technical**: open-source tools (Python 3.11+, PyTorch 2.6+, OpenCV, MediaPipe Tasks, Pygame) run on the TensorFlow Lite XNNPACK CPU delegate without proprietary drivers.
- **Operational**: standard plug-and-play webcams; the driver only looks at the road for the initial 15-second baseline routine.

- **Economic**: consumer laptops or low-cost SBCs (Raspberry Pi 5, Intel N100) reduce DMS hardware cost from thousands of dollars to under $50.

4.5 Selection of Edge Techniques and Toolchains

- **MediaPipe Tasks vs. dlib**: 478 3D landmarks in about 8 ms on CPU, against 68 landmarks in about 45 ms for dlib.
- **1D-CNN + GRU + TCN vs. 3D-ResNet**: video 3D-CNNs (SlowFast, I3D) need over 30 million parameters and high-end GPUs. Landmark-sequence modeling reduces the input to 12 channels and the parameter count to under 100k per model, enabling CPU execution.

CHAPTER 5 SYSTEM DESIGN AND METHODOLOGY

5.1 Pipeline Architectural Overview

ALERT follows a decoupled five-stage architecture (Fig. 5.1):

Stage	Description
1. Sensor Layer	Monocular RGB webcam capture (640×480 @ 15 FPS)
	↓
2. Landmark Extraction Layer	Google MediaPipe Face Landmarker producing 478 3D coordinates
	↓
3. Kinematic Feature Synthesis Layer	12-channel vector ($F = 12$) with personal baseline calibration
	↓
4. Heterogeneous Deep Ensemble Layer	120-frame (8.0 s) sequence inference across 9 models with soft-voting probability averaging
	↓
5. Debounced State Machine and HUD Layer	Online EMA smoothing driving NORMAL / WARNING / CRITICAL states, visual overlay and audio alarms

Fig. 5.1. ALERT end-to-end processing pipeline.

5.2 3D Facial Landmark Processing and Geometric Extraction

Landmarks are read from fixed anatomical indices:

- Left eye contour: [33,160,158,133,153,144]
- Right eye contour: [362,385,387,263,373,380]
- Outer mouth perimeter: [78,81,13,311,308,402,14,82]
- Head-pose anchors: [1 (nose), 152 (chin), 33,263,61,291]

EAR is computed per eye and averaged:

$$\mathrm{EAR}_{\mathrm{avg}} = \frac{1}{2}(\mathrm{EAR}_L + \mathrm{EAR}_R)$$

Yaw foreshortening is corrected geometrically when $|\psi| > 10°$:

$$\mathrm{EAR}_{\mathrm{corrected}} = \frac{\mathrm{EAR}_{\mathrm{avg}}}{\cos\psi}, \quad \cos\psi > 0.1$$

Vertical mouth opening is tracked by the Mouth Aspect Ratio:

$$\mathrm{MAR} = \frac{\| p_2 - p_8 \| + \| p_3 - p_7 \| + \| p_4 - p_6 \|}{3 \cdot \| p_1 - p_5 \|}$$

Head pitch (θ) and yaw (ψ) are recovered with cv2.solvePnP against a generic anthropometric 3D face model, with the focal length set to the frame width ($f_x = f_y = 640$).

5.3 Temporal Kinematic Features and Personal Baseline Calibration

5.3.1 The 12-Channel Kinematic Differential Vector

To separate dynamic gestures (speech, voluntary blinks) from fatigue (sluggish closures, yawning), the input is a 12-channel kinematic vector:

$$\mathbf{x}(t) = \begin{bmatrix} \text{EAR}_{L,\text{rel}} & \text{EAR}_{R,\text{rel}} & \text{EAR}_{\text{avg,rel}} & \Delta\text{EAR} \\ \text{MAR} & \Delta\text{MAR} & \text{PERCLOS}_{\text{calib}} & \text{Pitch}_{\text{rel}} \\ \Delta\text{Pitch} & \text{Yaw}_{\text{rel}} & \text{BlinkRate} & \text{BlinkDuration} \end{bmatrix}^{T} \in \mathbb{R}^{12}$$

Table 5.1: The 12 kinematic features synthesized per frame

Ch.	Feature	Definition	Diagnostic Property
F_1	$\text{EAR}_{L,\text{rel}}$	$\text{EAR}_L(t)/\text{EAR}_{\text{base}}$	Left eyelid aperture normalized to subject anatomy
F_2	$\text{EAR}_{R,\text{rel}}$	$\text{EAR}_R(t)/\text{EAR}_{\text{base}}$	Right eyelid aperture normalized to subject anatomy
F_3	$\text{EAR}_{\text{avg,rel}}$	$\text{EAR}_{\text{avg}}(t)/\text{EAR}_{\text{base}}$	Bilateral normalized aperture ($1.0 = $ open)
F_4	ΔEAR	$(\text{EAR}_t - \text{EAR}_{t-1})/\delta t$	Eyelid velocity; separates alert blinks from droops
F_5	MAR	Outer-lip aspect ratio	Vertical mouth opening
F_6	ΔMAR	$(\text{MAR}_t - \text{MAR}_{t-1})/\delta t$	Mouth expansion velocity; separates yawns from speech
F_7	$\text{PERCLOS}_{\text{calib}}$	Rolling fraction with $\text{EAR} < T_{\text{EAR}}$	Rolling eye-occlusion ratio
F_8	$\text{Pitch}_{\text{rel}}$	$\text{Pitch}(t) - \text{Pitch}_{\text{base}}$	Head nod angle relative to camera mount
F_9	ΔPitch	$(\text{Pitch}_t - \text{Pitch}_{t-1})/\delta t$	Nod velocity; detects micro-sleep drops
F_{10}	Yaw_{rel}	$\text{Yaw}(t) - \text{Yaw}_{\text{base}}$	Lateral gaze distraction
F_{11}	BlinkRate	Completed blinks per minute	Rolling blink frequency
F_{12}	BlinkDuration	Mean duration of recent blinks (ms)	Identifies sluggish drowsy blinks ($>$ 400 ms)

5.3.2 Personal Baseline Calibration and Population Spread

Resting baselines were extracted from each subject's Alert video (0.mov) across 48 subjects of UTA-RLDD:

- **Minimum resting EAR**: 0.1800 (Subjects 02, 07, 10)
- **Median resting EAR**: 0.3015
- **Maximum resting EAR**: 0.4500 (Subjects 04, 06, 11)
- **Ratio**: $0.4500/0.1800 = 2.50\times$

An awake Subject 02 ($\text{EAR} = 0.180$) is classified as 100% closed-eye drowsy by a static 0.25 threshold, whereas Subject 04 ($\text{EAR} = 0.450$) could half-close the eyes to 0.26 and still be classified as 100% alert.

ALERT's 15-second calibration computes:

$$\text{EAR}_{\text{base}} = \text{median}(\text{EAR}_{\text{clean}}), \quad \text{Pitch}_{\text{base}} = \text{median}(\text{Pitch}_{\text{clean}}), \quad \text{Yaw}_{\text{base}} = \text{median}(\text{Yaw}_{\text{clean}})$$

The personalized closure threshold and normalized coordinate are:

$$T_{\text{EAR}} = 0.75 \cdot \text{EAR}_{\text{base}}, \qquad \text{EAR}_{\text{rel}}(t) = \frac{\text{EAR}(t)}{\max(\text{EAR}_{\text{base}}, 0.15)}$$

All drivers are thereby mapped to a common frame in which 1.0 is open and ≤ 0.5 is closed.

[Insert figure: logs/plots/baseline_variance.png]

Fig. 5.2. Distribution of personal resting eye aspect ratios (EAR_{base}) across subjects in UTA-RLDD.

5.3.3 Calibrated PERCLOS and Receptive Horizon Expansion

Calibrated PERCLOS is the percentage of frames in a rolling 60-frame (4.0 s) buffer with $\text{EAR}_{\text{avg}} < T_{\text{EAR}}$.

The sequence window was expanded from 60 to **120 frames (**8.0 s @ 15 FPS**)**, which captures complete deep yawn cycles ($3.5 - 5.5$ s) and head-nod drop-and-startle recoveries ($2.0 - 4.5$ s) that 4-second windows clip and misclassify.

5.4 Fatigue Severity Levels and Two-Tier Debounced State Machine

ALERT maps model outputs to three biological classes: **Class 0** Alert (label 0), **Class 1** Low Vigilant (label 5) and **Class 2** Drowsy (label 10). Predictions are smoothed with an online Exponential Moving Average:

$$\widehat{\mathbf{P}}_t = 0.35 \cdot \bar{\mathbf{P}}_t + 0.65 \cdot \widehat{\mathbf{P}}_{t-1}$$

The smoothed vector drives a deterministic two-tier state machine with hysteresis (Table 5.2).

Table 5.2: Fatigue severity levels and debounced alarm specification

Level	State	Activation	Persistence	Action
L0	NORMAL (Alert)	$\hat{P}_{\text{Alert}} \geq 0.50$	1 frame	Solid green HUD; counters reset
L1	MONITORING	$\hat{P}_{\text{LowVig}} \geq 0.50$	< 3.0 s (< 45 frames)	Yellow indicator; silent tracking
L2	WARNING (Low Vigilant)	$\hat{P}_{\text{LowVig}} \geq 0.50$	≥ 3.0 s (45 frames)	Flashing yellow HUD; soft chime
L3	CRITICAL (Drowsy)	$\hat{P}_{\text{Drowsy}} \geq 0.45$	≥ 1.5 s (22 frames)	Pulsing red HUD; continuous alarm

5.5 Classification Architecture Suite

5.5.1 HybridCNNGRU (Temporal Conv + GRU)

HybridCNNGRU ($\approx$ 18.1k parameters) combines 1D temporal convolutions with a single-layer GRU:

- Input: $(B, 120, 12)$
- Conv1D block 1: Conv1D$(12 \to 32, k = 3)$ + BatchNorm + ReLU + Dropout(0.2)
- Conv1D block 2: Conv1D$(32 \to 64, k = 5)$ + BatchNorm + ReLU + MaxPool1D(2), giving $(B, 60, 64)$
- Recurrent stage: GRU$(\text{input} = 64, \text{hidden} = 48, \text{layers} = 1)$
- Temporal pooling: Bahdanau scalar attention over 60 timesteps, giving a 48-D context

- Head: Linear$(48 \rightarrow 32)$ + ReLU + Linear$(32 \rightarrow 3)$

5.5.2 TCNClassifier (Causal Dilated ResNet)

TCNClassifier ($\approx$ 96.5k parameters) uses five causal dilated residual blocks ($d \in \{1,2,4,8,16\}$, $k = 3$). The receptive field is $1 + 2 \cdot (3-1) \cdot (1+2+4+8+16) = 125$ frames, which exceeds the 120-frame window. The feed-forward design parallelizes across time and avoids vanishing recurrent gradients.

5.5.3 BiGRUAttention (Bidirectional GRU + Self-Attention)

BiGRUAttention ($\approx$ 38.0k parameters) uses a 2-layer bidirectional GRU (36 units per direction, 72 hidden units) with 4-head temporal self-attention to capture context on both sides of micro-sleep onsets.

5.5.4 Soft-Voting Ensemble Engine and Online EMA Smoothing

Predictions from all 9 models (3 architectures across 3 subject folds) are fused by soft-voting probability averaging:

$$\bar{\mathbf{P}}(c) = \frac{1}{9}\sum_{m=1}^{9} P_m\,(c \mid \mathbf{W})$$

This cancels the individual inductive weaknesses of each architecture.

5.6 Production Training Configuration and Ordinal Distance-Penalized Focal Loss

Standard Cross-Entropy treats all misclassifications symmetrically. We therefore combine Focal Loss [21] with an ordinal distance penalty:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Focal}} + \lambda_{\text{ord}} \cdot \mathcal{L}_{\text{Ordinal}}$$

$$\mathcal{L}_{\text{Focal}} = -\sum_{c=0}^{2} \alpha_c\,(1-p_c)^{\gamma} y_{c,\text{smooth}} \log(p_c), \quad \gamma = 2.0, \quad \epsilon = 0.08$$

$$\mathcal{L}_{\text{Ordinal}} = \sum_{c=0}^{2} p_c \cdot (c - y_{\text{true}})^2, \quad \lambda_{\text{ord}} = 0.35$$

Class weights are $\boldsymbol{\alpha} = [1.0, 1.15, 1.10]$.

Table 5.3: Production training hyperparameters

Hyperparameter	Value	Description
Dataset volume	119 verified videos, 26,050 windows	Genuine sleep-deprivation recordings (UTA-RLDD)
Split protocol	3-fold subject-independent GroupKFold	No subject identity leakage
Tensor dimension	$(B, 120, 12)$ (8.0 s @ 15 FPS)	12 kinematic channels
Optimizer	AdamW (lr $= 10^{-3}$, weight decay $= 10^{-3}$)	Decoupled weight decay
Schedule	CosineAnnealingLR ($T_{\max} = 50$, $\eta_{\min} = 10^{-6}$)	Smooth convergence to flat minima

Hyperparameter	Value	Description
Batch size and hardware	128; RTX 3050 6GB Laptop GPU, PyTorch AMP	Mixed precision (fp16)
Early stopping	Patience 15 epochs on validation Macro-F1	Limits late-stage over-fitting

CHAPTER 6 IMPLEMENTATION AND RESULTS

6.1 Implementation Environment and Video Decoding Acceleration

- **Host hardware**: Intel Core i7 / AMD Ryzen CPU, 16 GB RAM, NVIDIA GeForce RTX 3050 Laptop GPU (6 GB VRAM).
- **Edge inference target**: single-thread x86/ARM CPU, simulating a low-cost automotive ECU.
- **Software**: Linux (Ubuntu 22.04 LTS, Wayland and X11), Python 3.11+, PyTorch 2.6+ (CUDA 12.4, cuDNN), OpenCV 4.10+, MediaPipe Tasks 0.10+, Pygame 2.6.

Packet-level video acceleration (cap.grab()). Decoding 141 ten-minute videos (≈ 2.54 million frames) with standard read() took about 94 minutes. In scripts/extract_cache.py, discarded frames are advanced with cap.grab(), which moves the demuxer forward without YUV-to-BGR conversion, and only frames on the 15 FPS sampling interval are rasterized with cap.retrieve(). This gave an 8.9 × **speedup**, extracting a 10-minute video in under 12 s.

6.2 Modular Software Architecture

- src/core/feature_extractor.py: extracts 478 landmarks, computes EAR/MAR, resolves pose via solvePnP and synthesizes the 12 channels.
- src/core/calibrator.py: 15-second baseline calibration.
- src/core/model.py: HybridCNNGRU, TCNClassifier and BiGRUAttention.
- src/core/classifier.py: rolling deques, window normalizers, multi-model forward passes and soft-voting fusion.
- src/core/alert_manager.py: debounced two-tier state machine and audio chime synthesis.
- src/ui/dashboard.py and overlay.py: OpenCV HUD with an attention heatmap strip.
- run_webcam.sh: Linux launcher with automatic .venv discovery and camera fallback.

6.3 Empirical Evolution: Diagnostic Machine Learning Challenges

6.3.1 Challenge 1: Central Attractor Collapse (Phase 1)

A BiGRU trained with symmetric Cross-Entropy predicted Class 1 (Low Vigilant) for 86.0% of samples, giving a Macro-F1 of only **0.3428** (Table 6.1). Because Low Vigilant lies between Alert and Drowsy in latent space, the optimizer minimized risk by assigning the middle class to ambiguous samples.

Table 6.1: Phase 1 confusion matrix showing central attractor collapse (% of true class)

True State	Predicted Alert	Predicted Low Vigilant	Predicted Drowsy
Alert (Class 0)	7.1%	**66.2%**	26.8%
Low Vigilant (Class 1)	1.5%	**86.0%**	12.5%
Drowsy (Class 2)	0.4%	**68.5%**	31.1%

6.3.2 Challenge 2: Validation Loss Divergence (Phase 2)

In Phase 2, three candidate models evaluated on Fold 4 broke the attractor, but validation loss rose after epoch 2 and diverged until early stopping (Table 6.2).

Table 6.2: Phase 2 validation loss divergence (Fold 4 holdout), train / validation loss

Epoch	HybridCNNGRU	BiGRUAttention	TCNClassifier
1	0.6408 / 0.6270	0.6489 / 0.6277	0.6493 / 0.6312
2	0.6149 / **0.6142**	0.6174 / **0.6225**	0.6180 / **0.6254**
3	0.5931 / 0.6193 ↑	0.6086 / 0.6203 ↑	0.5982 / 0.6350 ↑
5	0.5649 / 0.6569 ↑	0.5891 / 0.6421 ↑	0.5614 / 0.6698 ↑
8	0.5380 / 0.7071 ↑	0.5752 / 0.6680 ↑	0.5391 / 0.7180 ↑
13	0.5113 / 0.7394 ↑	0.5676 / 0.6795 ↑	0.5116 / 0.7595 ↑

A forensic audit isolated four causes:

9. *Absolute EAR anatomical bias*: resting apertures varied from 0.18 to 0.45.
10. *Camera mount angle memorization*: pitch offsets of $\pm 15°$ were memorized as subject identity.
11. *Stride overlap (*96.7%*)*: stride-2 slicing allowed sequence autocorrelation to be memorized.
12. *Under-regularization*: softmax logits grew to ± 15, so ambiguous frames incurred severe penalties.

Figures 6.1 to 6.3 show stable convergence after the corrections.

[Insert figure: logs/plots/loss_curve.png]

Fig. 6.1. Training and validation loss under Ordinal Focal Loss, showing stable convergence without divergence.

[Insert figure: logs/plots/accuracy_curve.png]

Fig. 6.2. Holdout validation accuracy across training epochs.

[Insert figure: logs/plots/f1_curves.png]

Fig. 6.3. Per-class F1-score across training epochs.

6.3.3 Challenge 3: Live Webcam Train-Serve Skew

In live tests of the Phase 3 models, awake users with wide-open eyes were persistently classified as Low Vigilant. Two bugs were found:

13. **Window P90 blindness**: dividing EAR by the window's 90th percentile erased absolute scale, so awake eyes ($\text{EAR} = 0.30$) and drowsy eyes ($\text{EAR} = 0.18$) both produced an input of 1.0.
14. **PERCLOS train-serve skew**: cache extraction used an uncalibrated static threshold (0.25), giving a training PERCLOS mean of **60.99%** (std 0.4568). Live calibration produced a true PERCLOS of 0.0%, a normalized input of $z = (0.0 - 0.6099)/0.4568 = -1.335$, which pushed the model out of distribution into its Low-Vigilant bias.

Table 6.3: Cross-validation summary across four subject folds (Phase 3 BiGRU)

Fold	Accuracy	Macro-F1	Alert F1	LowVig F1	Drowsy F1	Macro-AUC	Best Val Loss
Fold 1	40.52%	0.4024	0.4196	0.4119	0.3757	0.5968	0.5367
Fold 2	45.41%	0.4495	0.4058	0.4480	0.4948	0.6009	0.5622
Fold 3	45.68%	0.4584	0.4398	0.4530	0.4823	0.6134	0.5680
Fold 4	43.28%	0.4274	0.3812	0.4405	0.4917	0.6247	0.5712
Mean ± Std	**43.72%**	**0.4344 ± 0.02**	**0.4116**	**0.4384**	**0.4611**	**0.6090**	**0.5595**

6.4 Experimental Results on UTA-RLDD

6.4.1 9-Model Ensemble Holdout Leaderboard

With the 12 kinematic channels, personal baseline calibration and Ordinal Distance-Penalized Focal Loss, all failure modes were resolved. Table 6.4 reports the 9-model ensemble on unseen holdout subjects across the three folds. Checkpoints follow the pattern ensemble_<architecture>_fold<k>.pt.

Table 6.4: 9-model ensemble holdout leaderboard (subject-independent holdout folds)

Architecture	Fold	Best Epoch	Macro-F1	Accuracy	$F1_{\text{Alert}}$	$F1_{\text{LowVi}}$	$F1_{\text{Drowsy}}$	Val Loss	GPU Time
CNN-GRU	1	1	0.5252	52.54%	0.4801	0.5076	0.5878	0.9396	102.9 s
CNN-GRU	2	4	0.4945	49.89%	0.4397	0.4466	0.5973	1.3445	142.5 s
CNN-GRU	3	2	**0.6062**	**60.37%**	**0.6622**	**0.5118**	0.6445	**0.8066**	114.9 s
TCN	1	4	0.4986	49.51%	0.5502	0.3989	0.5468	1.7938	173.5 s
TCN	2	1	0.4872	50.46%	0.4122	0.3870	**0.6623**	1.2091	129.9 s
TCN	3	1	0.5430	54.52%	0.6407	0.4367	0.5516	0.9123	132.1 s
BiGRU	1	2	0.5361	53.43%	0.5161	0.5132	0.5789	1.2444	116.7 s
BiGRU	2	6	0.4741	47.67%	0.4486	0.4021	0.5715	1.7927	161.1 s
BiGRU	3	10	0.5472	54.63%	0.5826	0.4311	0.6280	1.7038	209.3 s
Ensemble average	–	–	**0.5236**	**52.56%**	**0.5203**	**0.4483**	**0.5965**	**1.3052**	**19.7 min total**

[Insert figure: logs/plots/confusion_matrix.png]

Fig. 6.4. Normalized holdout confusion matrix, showing strong diagonal sensitivity and suppression of Alert-vs-Drowsy opposite-extreme errors.

[Insert figure: logs/plots/architecture_comparison.png]

Fig. 6.5. Architectural benchmark radar comparing Macro-F1, per-class F1 and parameter footprint.

6.4.2 Master Model Benchmark (All 141 Videos)

Trained on all 141 videos (models/phase3/master_model.pt), the master model achieves:

- **Accuracy**: 65.61%
- **Macro-F1**: 0.6600; **Weighted F1**: 0.6589; **Macro ROC-AUC**: 0.8413
- **Alert**: precision 70.87%, recall 67.71%, F1 0.6926, AUC 0.8630
- **Low Vigilant**: precision 55.45%, recall 67.16%, F1 0.6075, AUC 0.7512
- **Drowsy**: precision 76.13%, recall 61.42%, F1 0.6799, AUC 0.9099

6.5 Hardware Latency Profiling and Edge Viability

Benchmarks ran 1,000 continuous cycles ($T = 120$, $F = 12$, $B = 1$) on a single-thread Intel host CPU and an NVIDIA RTX 3050 Laptop GPU (Table 6.5).

Table 6.5: Latency, parameter and throughput benchmark

Model	Parameters	Checkpoint	CPU Mean	CPU p95	CPU FPS	GPU Mean	Edge Suitability
HybridCNNGRU	18,131	0.069 MB	15.30 ms	30.70 ms	65.3	4.56 ms	Strict edge compliant
BiGRUAttention	37,971	0.145 MB	42.04 ms	55.05 ms	23.8	2.91 ms	Soft real-time
TCNClassifier	96,499	0.368 MB	51.43 ms	79.74 ms	19.4	12.36 ms	GPU recommended
Full 9-model ensemble	610,404	2.35 MB	511.65 ms	607.98 ms	2.0	80.73 ms	GPU real-time (12.4 FPS); not real-time on CPU

[Insert figure: logs/plots/latency_pareto.png]

Fig. 6.6. Inference latency vs. Macro-F1 on edge CPU and GPU hardware.

HybridCNNGRU runs in **15.30 ms** on CPU and uses only **7.2%** of the 250k parameter budget while reaching up to **0.6062** Macro-F1, making it the best standalone model for automotive ECUs.

CHAPTER 7 TESTING

7.1 Verification Strategy

1. **Automated unit and integration tests**: tensor shapes, normalizer fitted state, landmark indexing and model forward-pass stability.
2. **Stress and error-state tests**: fallback when Linux video device indices vary (/dev/video0 vs. /dev/video1), missing-checkpoint remapping and audio mixer fallbacks.
3. **Live webcam tests**: closed-loop checks that awake drivers produce no false alarms and that sustained eye closure triggers audio-visual alarms within 22 frames.

7.2 Integration and Unit Test Cases (57/57 Automated Suite)

Table 7.1: Integration and regression test cases

ID	Module	Condition	Expected Result	Status
T01	feature_extractor	Normalized landmark array ingestion	Shape $(12,)$ float32 returned	PASS
T02	feature_extractor	Head yaw $> 10°$	Cosine foreshortening correction applied	PASS
T03	calibrator	15-second landmark stream	$\mathrm{EAR}_{\mathrm{base}}$ and $\mathrm{Pitch}_{\mathrm{base}}$ profile saved	PASS
T04	calibrator	Rapid blink frames ($\mathrm{EAR} < 0.12$)	Discarded from clean baseline median	PASS
T05	model	HybridCNNGRU forward pass $(1,120,12)$	Logits $(1,3)$ and attention $(1,60)$	PASS
T06	model	TCNClassifier forward pass $(1,120,12)$	Logits $(1,3)$ and dummy attention	PASS
T07	model	BiGRUAttention legacy checkpoint load	Remaps fc1 to head.0 without crash	PASS
T08	classifier	EnsembleFatigueClassifier initialization	Loads all 9 models and companion .npz	PASS
T09	classifier	Online EMA smoothing update	Smooths probability spikes ($\alpha = 0.35$)	PASS
T10	alert_manager	Sustained Low Vigilant (45 frames)	Transitions to yellow Warning	PASS
T11	alert_manager	Sustained Drowsy (22 frames)	Transitions to red Critical with siren	PASS
T12	dashboard	/dev/video0 metadata failure	Probes and falls back to /dev/video1	PASS
T13	frame_extractor	_LandmarkAdapter subscript indexing	adapted[i] accesses landmark directly	PASS
T14	run_alert.py	CLI --ensemble flag	Launches 9-model ensemble mode	PASS
T15	Regression suite	Complete integration suite	57 / 57 tests passing in 2.75 s	PASS

7.3 Live Webcam Stress-Testing Outcome

Run on a live monocular webcam stream (./run_webcam.sh):

4. **Calibration**: the driver keeps natural open eyes for 15 seconds while an amber progress bar fills, capturing $\mathrm{EAR}_{\mathrm{base}} \approx 0.315$ and $\mathrm{Pitch}_{\mathrm{base}} \approx -2.1°$.
5. **Awake driver**: looking ahead naturally gives $\hat{P}(\mathrm{Alert}) \in [0.85, 0.96]$ and a solid green NORMAL (Alert) HUD, resolving the earlier false-alarm bias.

6. **Prolonged eye closure**: closing the eyes for more than 1.5 s turns the HUD to flashing red CRITICAL (Drowsy) and starts a continuous siren within 22 frames (1.46 s).

CHAPTER 8 CONCLUSION AND DISCUSSION

8.1 Summary of Project Work

ALERT delivers an edge-compliant, fully offline driver fatigue detector that addresses the main failure modes of prior DMS designs. A 120-frame (8.0 s) event horizon, a 12-channel kinematic vector with velocity differentials, personal baseline calibration and an Ordinal Distance-Penalized Focal Loss remove the central attractor collapse and the bias from individual eye morphology. Under strict subject-independent cross-validation on UTA-RLDD, the 9-model ensemble reaches a peak holdout Macro-F1 of **0.6062** and an average of **0.5236**. The deployable HybridCNNGRU runs at **15.30 ms** on a single CPU thread, and the full ensemble occupies **2.35 MB** on disk.

8.2 Problems Encountered and Resolved

15. *Central attractor collapse*: resolved by the Ordinal Distance-Penalized Focal Loss, which penalizes opposite-extreme errors 4 × harder.
16. *Validation loss divergence*: resolved by personal baseline scaling, head-pose median centering, zero-overlap training slicing and AdamW regularization.
17. *Window P90 blindness and PERCLOS skew*: resolved by replacing window-relative division with true personal baselines (EAR/EAR_{base}), which removed live false alarms.
18. *Linux camera device indexing*: automatic probe fallback in dashboard.py from metadata node 0 to capture node 1.
19. *Subscriptable adapter indexing*: sequence methods added to _LandmarkAdapter for direct indexing during calibration.

8.3 Limitations of Current System

20. *Ocular occlusion*: reflective eyewear or opaque sunglasses obstruct eye tracking.
21. *Low light*: standard RGB cameras degrade in unlit night driving without near-infrared (NIR) illumination.
22. *Single-camera view angle*: head turns beyond 45° cause landmark tracking loss.

8.4 Future Research and Industrial Deployment Roadmap

1. **NIR sensor integration**: 850 nm / 940 nm active illuminators with CLAHE image enhancement [20] for dark conditions.
2. **Automotive SoC compilation**: export to ONNX Opset 14 and INT8 quantization via TensorRT/OpenVINO for automotive controllers (TI TDA4VM, Ambarella CV2x, NXP S32G).
3. **CAN-bus haptic actuation**: connect critical alarm triggers to automotive CAN networks to drive seat vibration motors and active seatbelt tensioners.
4. networks to drive seat vibration motors and active seatbelt tensioners.

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